1. In a perfectly competitive market that is in a long run equilibrium, price will be equal to (MARK ALL THAT APPLY)

a) Marginal Revenue
b) Marginal Cost
c) Average Variable Cost
d) Fixed Cost
e) None of the above

2. Jon's bike shop operates in a perfectly competitive market where the prevailing price for a bike is $300. If Yann's marginal cost function is given by MC=0.5q. If fixed costs are $100,000, then profit is _____and the firm _____ in the short run.

a) Positive, continue to operate
b) Positive, shutdown
c) Zero, shutdown
d) Negative,continue to operate
e) Negative, shutdown

3. Jon's E-bike shop operates in a perfectly competitive market where the prevailing price for a bike is $900. If Yann's marginal cost function is given by MC=0.5q. If fixed costs are $100,000, all else being equal, what do you expect to happen to the price in this market due to perfect competition in the long run? (MARK ALL THAT APPLY)

a) It will increase
b) It will decrease
c) It will equal Marginal Cost
d) It will be less than the current marginal cost
e) It will be less than average total cost

4. A firm is operating in a perfectly competitive market and making positive economic profit. Which of the following accurately describes the relationship between Marginal Cost and Average Cost?

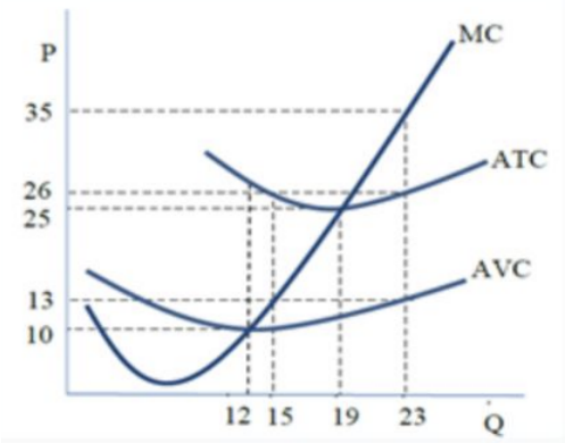
a) MC>AC
b) MC<AC
c) MC=AC
d) Need more information
e) The relationship between MC and AC is best described as giving the long distance thing a try, but will probably break up over thanksgiving break.

5. The opportunity cost of a business owner's time would be included in–

a) Accounting Profit
b) Economic Profit
c) Both accounting and economic Profit
d) Neither Accounting nor economic profit

6. There are 10 identical firms operating in a perfectly competitive market, each with cost function: $C(q)=300+q^{2,}$ and market demand is Q=60-P. What is the price at the equilibrium?

A) 1250
B) 250
C) 200
D) 10
E) None of the above



7. Which of the following prices would have the lowest profit for the firm in the above market (assuming perfect competition)?

a) $35
b) $26
c) $25
d) $13
e) $1

8. There are 10 identical firms operating in a perfectly competitive market, each with cost function: $C(q)=300+q^{2,}$ and Market demand is Q=60-P. What is deadweight loss at the equilibrium?

a) 1250
b) 250
c) 200
d) 10
e) None of the above

9. A firm is operating in a perfectly competitive market, price is $0.50 and at the optimal quantity, average cost is $0.75. In the long run the market supply curve will ____ and price will ____

a) increase, increase
b) increase, decrease
c) decrease, increase
d) decrease, decrease
e) need more information

10. A firm operates in a perfectly competitive market where the prevailing price for their product is $20. The marginal cost function is given by MC=2q. If fixed costs are $2,000, in the short run the firm will (MARK ALL THAT APPLY)

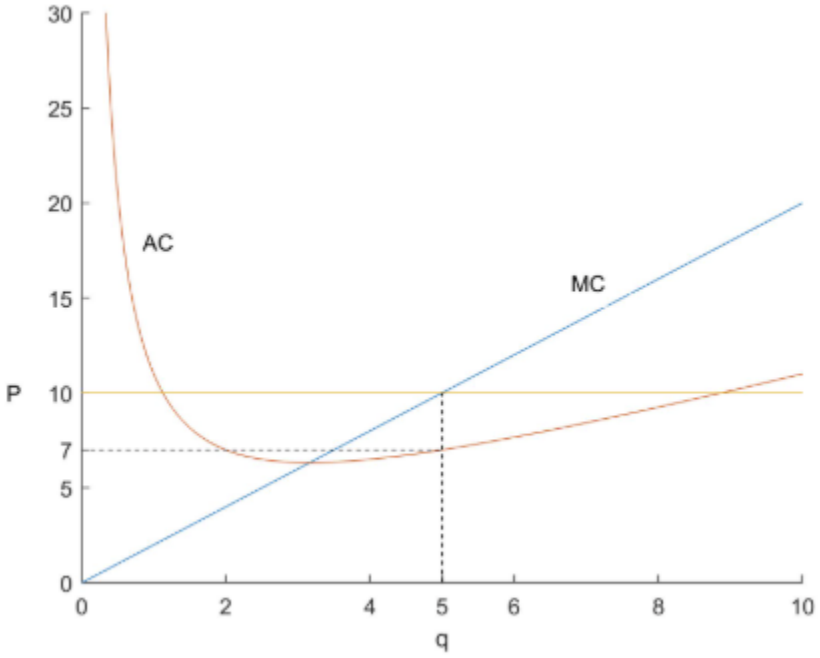
a) Continue to operate
b) Make negative profit
c) Produce 10 units
d) Produce 0 units
e) Have positive variable profit

11. If I invest $100,000 in my business and the rate of return is 5% if I would have invested it elsewhere, the opportunity cost of a capital invested in a business would be (MARK ALL THAT APPLY)–

a) $100,000
b) $95,000
c) $5,000
d) Included in economic profit
e) Included in accounting profit

12. A firm is operating in a perfectly competitive market, price is $6 and at the optimal quantity, average cost is $9. In the long run the market supply curve will ____ and price will ____

a) Increase, increase
b) Increase, decrease
c) Decrease, increase
d) Decrease, decrease
e) Need to know average variable cost to answer the question



Consider the above graph for the next two questions

13. At the price of $10, firms will make a total profit of–
a) 15
b) 25
c) 10
d) 5
e) Negative profit

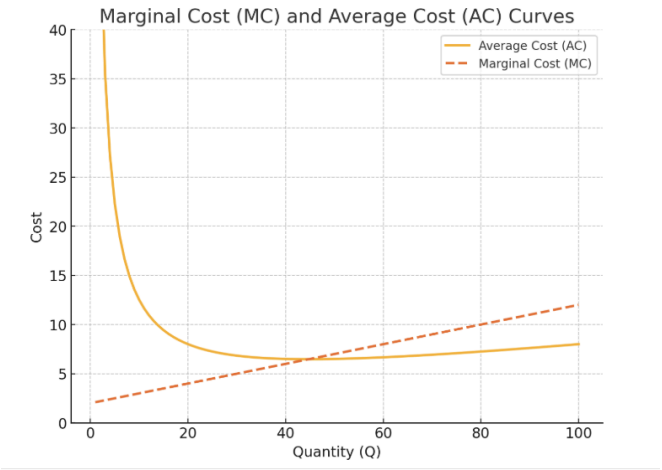
14. In the long run, the price will be driven to (assume perfect competition)–
a) More than $10
b) $10
c) $7
d) Less than $7
e) None of the above

15. Jimmy Johnson's fruit stand operates in a perfectly competitive market where the prevailing price for a coconut (his only product) is $3. Suppose Jimmy's marginal cost function is given by MC=0.1q. For which level of fixed costs should he operate? (MARK ALL THAT APPLY)

a) FC=0
b) FC=10
c) FC=50
d) FC= 100
e) FC=1000

16. There are 80 identical firms, each with an individual supply curve q=0.5p. If the market price is P=8, what is total producer surplus at the perfectly competitive equilibrium?

A. 640
B. 1,280
C. 2,560
D. 5,120
E. None of the above

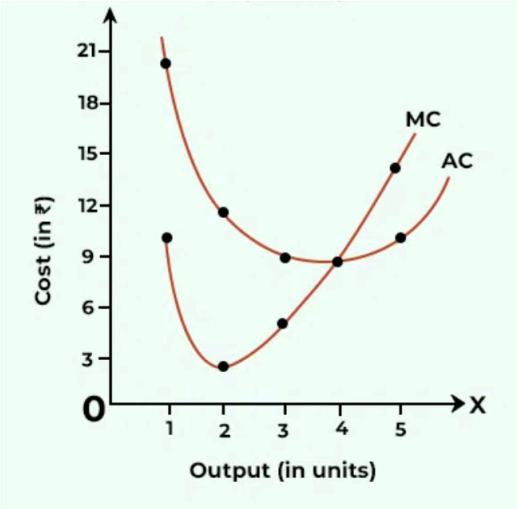


17. Use the above graph. The price in the above perfectly competitive market is $5. Which of the following statements is true (MARK ALL THAT APPLY)

a) The market is in a long run equilibrium
b) The firm will shutdown at that price
c) In the long run, the price will rise
d) The firm is making negative variable profit.
e) None of the above

18. Suppose the cost function for a firm is given by C(q) = 400 + 5q. Fixed cost is _____ and marginal cost is _____

a) 400, 5
b) 400, 5q
c) 5, 400
d) 0, $5q^2$
e) None of the above

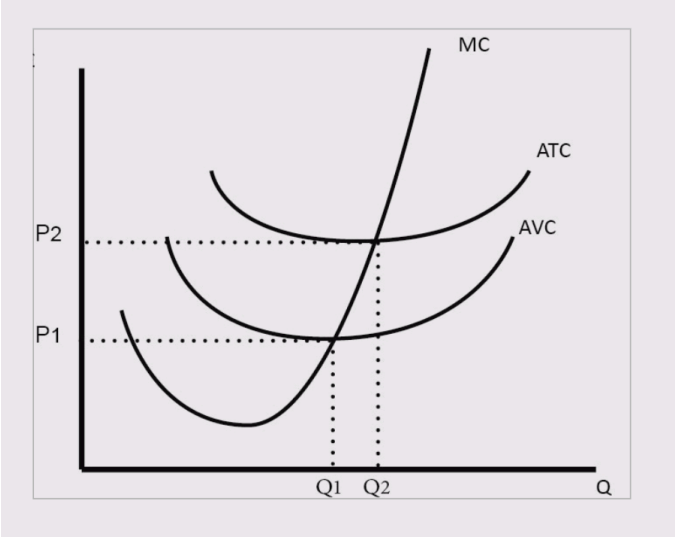


19. Refer to the above graph. Assume the curves represent the Marginal and Average Cost curves for a firm in a perfectly competitive market.
Which of the following statements is true and the long run equilibrium price in this market? (MARK ALL THAT APPLY)

a) the long run price is lower than $15
b) The firm will exit in the long run
c) The long run price is higher than $6
d) The long price is where the Marginal cost curve crosses the average cost curve
e) At the long run equilibrium, the firm will earn zero profit

20. There are 10 identical firms operating in a perfectly competitive market, each with cost function: $C(q)=100+q^2$ and Market demand is Q=60-P. What is consumer surplus at the equilibrium?

a) 1250
b) 250
c) 200
d) 10
e) None of the above



21. Refer to the above graph, for what prices does the firm earn negative profit? (MARK ALL THAT APPLY)

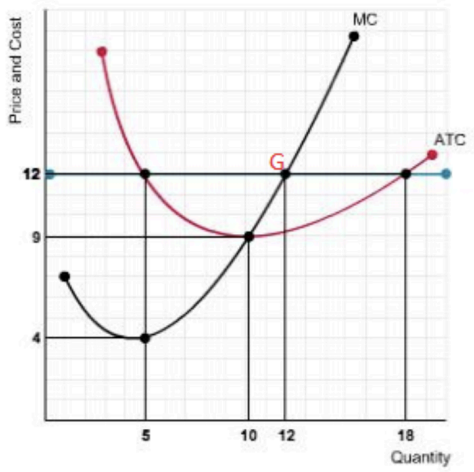
a) Less than P_1
b) Between P_1 and P_2
c) Greater than P_2
d) None, the firm will shut down if profit was negative and earn zero profit instead

22. There are currently 60 perfectly competitive firms producing output q. Their cost function is C=10+3q2 . The demand in this market is QD=1200-40p. Total producer surplus is?

a) 48
b) 2880
c) 24
d) 240
e) 32

23. A firm operates in a perfectly competitive market where the prevailing price for their product is $20. The marginal cost function is given by MC=5q. If fixed costs are $100, in the short run the firm will (MARK ALL THAT APPLY)

a) Continue to operate
b) Make negative profit
c) Produce 4 units
d) Produce 0 units
e) Have positive variable profit



24.

Refer to the above graph of a firm's cost curves for the next two questions. Assume the market is in perfect competition. If the market price is $12, how much will the firm produce?

a) 18 b) 12 c) 10 d) 5 e) 0

25. In the long run, what will be the price in this market?

a) 18
b) 12
c) 9
d) 4
e) 0

26. There are 10 identical firms operating in a perfectly competitive market, each with cost function: $C(q)=100+q^2$ and Market demand is $Q=60-P$. What is total surplus at the equilibrium?

a) 1500
b) 1250
c) 250
d) 150
e) 10

27. Use this info for the next two questions. There are 4 identical firms operating in a perfectly competitive market, each with cost function:

$C(q)=25+q^2$

Market demand is:

$Q=30-P$

28. What is the total surplus at the competitive equilibrium?

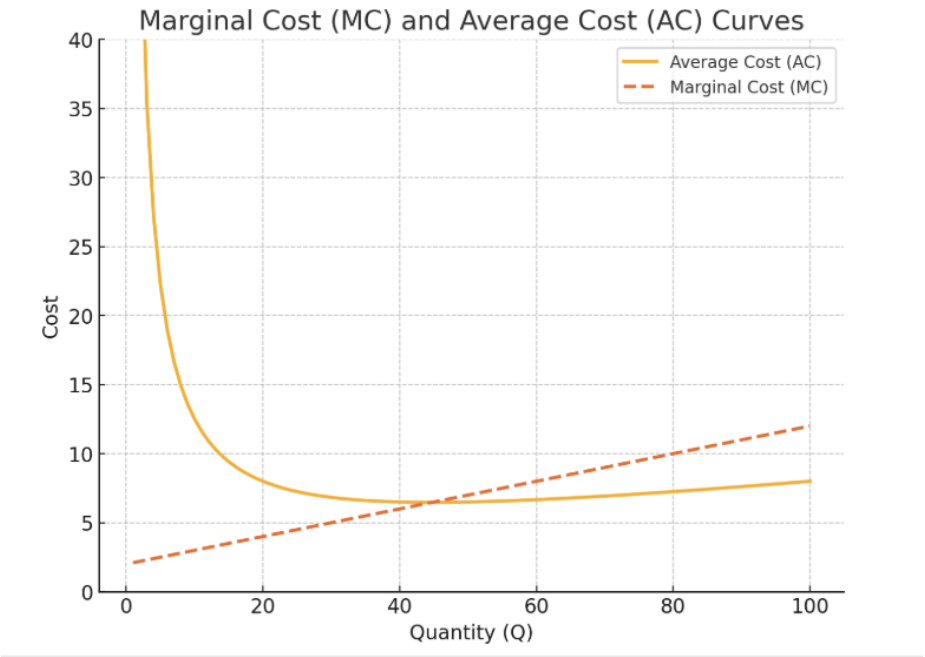
a) 300
b) 200
c) 150
d) 100
e) 75

29. In the above question, what do you expect to happen to the number of firms in the long run?

a) There will be more than 4
b) There will be less than 4
c) There will be 4
d) Depends on the fixed cost, which is not given

30. A firm has variable profit of $10, while its fixed costs are $50. Should the firm shut down?

a) Yes
b) No
c) Not enough information
d) The above scenario is impossible



31. Use the above graph. The price in the above perfectly competitive market is $5. Which of the following statements is true (MARK ALL THAT APPLY)

a) The market is in a long run equilibrium
b) The firm will shutdown at that price
c) In the long run, the price will rise
d) The firm is making negative variable profit.
e) None of the above

32. There are 10 identical firms operating in a perfectly competitive market, each with cost function: $C(q)=300+q^2$ and Market demand is $Q=60-P$. What is consumer surplus at the equilibrium?

a) 1250
b) 250
c) 200
d) 10
e) None of the above

33. A firm is operating in a perfectly competitive market, price is $1.00 and at the optimal quantity, average cost is also $1.00. In the long run the market supply curve will ___ and price will ___

a) increase, increase
b) increase, decrease
c) decrease, increase
d) decrease, decrease
e) None of the above

34. In a perfectly competitive market, there are 8 identical firms, each with a cost function: $C(q) = 200 + 2q^2$. The market demand is given by $Q = 100 - P$. What is the equilibrium price?
A) $20
B) $40
C) $60
D) $80
E) None of the above

35. If I invest $50,000 in my business and the rate of return is 10% if I would have invested it elsewhere, the opportunity cost of capital invested in a business would be (MARK ALL THAT APPLY)–

a) $50,000
b) $10,000
c) $5,000
d) Included in economic profit
e) Included in accounting profit